

Hole trapping and emission behavior near the valence band edge at an AlSiO/p-type GaN interface revealed by repeated temperature-dependent C-V measurements

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ABSTRACT

Bias instability associated with traps near the valence band edge at gate dielectric/p-type GaN interfaces poses a serious reliability concern for GaN metal-oxide-semiconductor field-effect transistors although origins of hole traps are still unclear. This study demonstrates an isolation of trap components with interface states and near-interface traps (NITs) at an AlSiO/p-type GaN interface by analyzing the hole emission process as a function of temperature and sweep time. The gate dielectric was a 40 nm-thick AlSiO film deposited by plasma-enhanced atomic layer deposition at 250 °C and annealed at 400 °C in nitrogen gas. The capacitance-voltage (C-V) characteristics exhibited a significant plateau and no hole accumulation in the valence band of GaN, indicating a high density of interface traps at approximately 0.8 eV above the valence band maximum of GaN. After the first reverse C-V sweep, the subsequent forward sweep showed a large negative shift (hysteresis) that largely persisted in the following reverse sweeps. At elevated temperatures and with slower sweep rates, the shift was partially reduced (recovered); however, a residual shift remained. An Arrhenius analysis of the recovered component yielded an activation energy of approximately 0.8 eV, which was consistent with the Fermi-level pinning position, suggesting that the thermally recovered component is governed by hole emission from interface states. In contrast, the persistent residual shift suggested a negligibly emitting component, i.e., NITs. We, thus, isolated the contribution of interface states from NITs in the hole emission process.

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Gallium nitride (GaN)-based metal-oxide-semiconductor field-effect transistors (MOSFETs) are expected to be adopted as next-generation power switching devices.¹ To that end, the formation of a high-quality MOS interface is essential to achieve the expected excellent performance of GaN MOSFETs. Achieving a high-quality interface requires suitable selection of the gate dielectric and the establishment of optimized processing procedures. In particular, enough band offsets relative to GaN and a high relative permittivity are demanded for the gate dielectric of a GaN MOSFET to suppress gate leakage current and to achieve a high transconductance, respectively.² Silicon dioxide (SiO₂),^{3–15} aluminum oxide (Al₂O₃),^{16–25} and their mixed oxides (AlSiO)^{26–33} have been investigated based on these considerations. GaN MOSFETs using these dielectrics have exhibited electron mobilities exceeding 100 cm² V⁻¹ s⁻¹.^{10,15,25,27}

In this study, we selected AlSiO as a gate dielectric. AlSiO was selected as the gate dielectric because it offers a favorable balance of leakage suppression, interface quality, and reliability for GaN MOS devices.³⁴ In addition, first-principles calculations suggested that AlSiO/GaN interfaces can be free from dangling-bond-related deep gap states.³⁵ Previous studies on AlSiO/n-type GaN MOS capacitors have shown that post-deposition annealing (PDA) suppresses hysteresis and flatband shifts and extends dielectric lifetime, which relate to electron trapping and/or injection.²⁶ For GaN MOSFETs, a thin crystalline AlN interlayer at the AlSiO/p-type GaN interface has been shown to increase channel mobility.^{36–38} Conversely, analyses of lateral MOSFETs have indicated high-density hole trapping under negative gate bias,³⁰ which leads to a negative bias instability in threshold voltages. Similar hole trapping under a negative gate bias has also

been observed with SiO₂ gate dielectrics and has been attributed to traps near the valence band maximum (E_V) within GaO_x formed at the SiO₂/p-type GaN interface.³ Moreover, the thickness of the GaO_x depends on the temperature during the oxide formation process; lower deposition temperatures and subsequent post-treatments tend to reduce its thickness.³⁹

At an AlSiO/GaN system, a good interface property near the conduction band has been reported,³² whereas a high density of interface traps has been suggested near the E_V .³⁰ The hole traps at an AlSiO/p-type GaN interface would involve both interface states and near-interface traps (NITs). Tomigahara *et al.*⁵ suggest two components of hole traps based on the hump and hysteresis in the C-V curves of SiO₂/p-type GaN MOS capacitors. On the basis of their explanation, the hump is attributed to Fermi-level pinning caused by a high density of interface states.⁵ Once interface states are filled in a large negative bias, holes start to be trapped to oxide traps (i.e., NITs) in the further negative sweep, and the subsequent forward sweep is shifted to the negative direction, resulting in causing a large C-V hysteresis.⁵ However, previous studies usually characterized hole traps at room temperature, resulting in limited hole emission from traps with deep energy levels. Holes trapped in deep interface states are expected to be thermally emitted with a long time constant, while holes in NITs are hardly emitted due to the spatial separation from the interface. Therefore, an analysis of hole emission processes at elevating temperatures over long times is a key to accurately separate components of hole trapping. **The present study performs analyses of temperature-dependent C-V measurements with multiple sweeps at various sweep rates.** To the best of our knowledge, it is the first demonstration of clear thermal emission from deep hole traps at oxide/p-type GaN interface. The results clearly demonstrate a separation of components by interface states and NITs in an AlSiO/p-type GaN system.

As a substrate, we used a freestanding GaN wafer prepared by hydride vapor phase epitaxy. The layered structure grown by metalorganic vapor phase epitaxy was composed of a 2 μm -thick n^+ -type GaN layer doped with a Si concentration at $3 \times 10^{18} \text{ cm}^{-3}$, a 1 μm -thick p^+ -type GaN doped with Mg at $3 \times 10^{18} \text{ cm}^{-3}$, and a 1 μm -thick

p-type GaN with Mg at $5 \times 10^{17} \text{ cm}^{-3}$, in the bottom to the top order. Because the capacitance of the bottom p^+/n^+ junction was much greater than that of the gate capacitance, the observed capacitance was almost attributed to be the gate capacitance.²⁴ As discussed later with Fig. 1(b), the dual sweep from +5 to -4 V and subsequently from -4 to +5 V showed no clear hysteresis, suggesting that charge-storage effects associated with the bottom p^+/n^+ diode and the floating p-type GaN layer were negligible. A 40 nm-thick AlSiO gate dielectric was deposited by plasma-enhanced atomic layer deposition at 250 °C. The Si/(Si + Al) ratio in AlSiO was set to 22%.²⁹ Prior to dielectric deposition, an *in situ* N₂ radical surface treatment was performed. A 3 nm-thick SiO₂ capping layer was deposited on the AlSiO layer. Post-deposition annealing (PDA) was then carried out at 400 °C for 30 min in a N₂ ambient. Tomigahara *et al.*⁵ reported that PDA temperature more than 600 °C additionally generated hole traps in an SiO₂/p-type GaN system. The present paper focuses on only results for PDA at 400 °C, and the effects of PDA temperature for an AlSiO/p-type GaN system will be discussed elsewhere. Gate electrodes with a 400 μm diameter were formed by depositing Al via resistance heating evaporation. For the backside contact, Ti/Al/Ni/Au with thicknesses of 20/200/30/30 nm was deposited by electron-beam evaporation. The structure of the fabricated MOS capacitor is shown in Fig. 1(a).

Capacitance-voltage (C-V) measurements were performed on the fabricated MOS capacitors at a frequency of 1 kHz. The DC bias was swept from positive to negative voltages (reverse sweep) and then returned from negative to positive voltages (subsequent forward sweep). The resulting C-V characteristics are summarized in Fig. 1(b). To exclude measurement-induced persistent shifts, fresh electrodes were used for respective curves. As shown by the blue line in Fig. 1(b) (from 5 to -15 V), the capacitance during the reverse sweep remained almost constant over a wide range (from -5 to -15 V) on the negative bias side. As reported previously for SiO₂/p-type GaN MOS capacitors,⁵ this behavior is attributed to Fermi-level pinning caused by interface traps. The dashed line in Fig. 1(b) represents the ideal C-V curve, assuming hole accumulation in the E_V of GaN. On the basis of the relationship between capacitance and surface potential in the ideal curve, the experimental plateau capacitance is

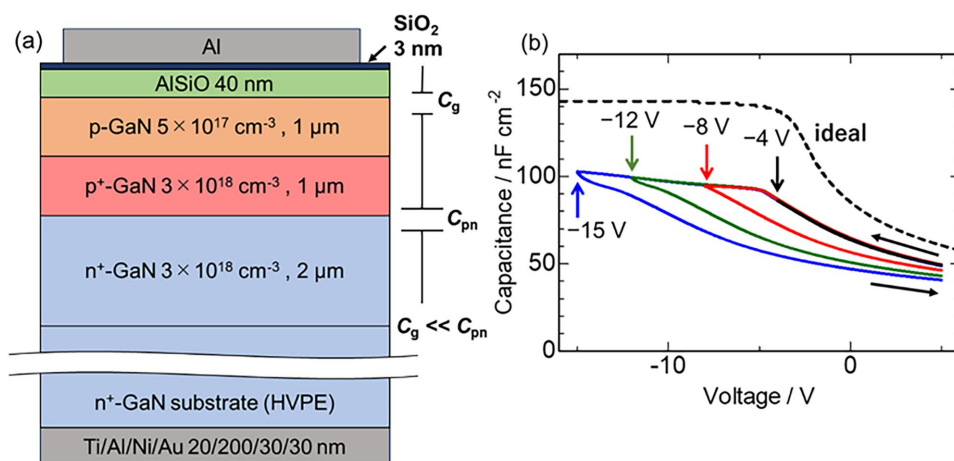


FIG. 1. (a) Cross-sectional diagram of the MOS capacitor structure with the capacitance equivalent circuit. (b) C-V characteristics measured at 30 °C for sweep endpoints of -4, -8, -12, and -15 V (sweep sequence: +5 V $\rightarrow$ negative end point $\rightarrow$ +5 V). The sweep rate was set to 0.5 V/s.

estimated to correspond to a surface pinning level at approximately $E_V + 0.8$ eV. It should be noted that this pinning level is only an estimation based on the 1 kHz C-V characteristics, rather than a rigorous analysis. Within the present measurement range (down to -15 V), the surface Fermi level remained pinning and was much lower than the ideal accumulation capacitance of 144 nF/cm². The subsequent forward sweep from -15 to 5 V was shifted negatively, resulting in the large hysteresis in the C-V curve.

To investigate the Fermi-level pinning in various sweep ranges, the negative bias end points were varied from -4 to -15 V. When the negative bias end point was set to -4 V below the Fermi-level pinning voltage, no hysteresis was observed as shown by the black solid line in Fig. 1(b). On the other hand, the capacitance plateau and hysteresis were observed when the negative end points were less than -8 V. The hysteresis was enhanced by extending negative bias range, indicating that trapped hole density was increased. The hole trap density in the pinning region was approximately estimated using $N_{\text{hump}} = C_{\text{OX}}\Delta V_{\text{hump}}/q$.⁵ Assuming a voltage range from -4 to -15 V, the density was estimated to be 1×10^{13} cm⁻². Since the Fermi level remained pinned at -15 V, the actual trap density may be even higher.

Figure 2(a) shows the results of the C-V measurements when the voltage sweep rate was varied from 0.02 to 4 V/s at 30°C . During the reverse sweep, the measured capacitance slightly increased with increasing the sweep rate in the -10 to -15 V range as highlighted by a red dotted circle. However, changes in capacitances were small, and magnitudes of the hysteresis were comparable for different sweep rates. These observations indicate that holes are quickly trapped but need the emission time longer than the present measurement timescale.

Figure 2(b) shows the results of the C-V measurements with a sweep rate of 0.5 V/s for various measurement temperatures from 30 to 200°C . During the reverse sweep, the measured capacitance increased slightly near -15 V with increasing the temperature as highlighted by a dot circle in Fig. 2(b). The slight increase in capacitance at higher temperature indicates partial hole emission during measurement time.

Next, repeated C-V measurements were carried out on the same MOS capacitors without any intervening recovery time. In these measurements, both the measurement temperature and the sweep rate

were varied. Figure 3 shows four representative plots, including measurements at 30 and 200°C taken with sweep rates of 0.5 and 0.02 V/s. In the initial dual sweep, the large hysteresis was observed as already discussed. The subsequent second reverse sweep was close to the initial forward sweep because most of holes remained trapped. By repeating sweeps, the negative shift was saturated and almost stabilized in the fifth sweep (see the supplementary material). In other words, hole trapping process was considered to be saturated in the fifth sweep. We thus only display the first and fifth sweeps in Fig. 3. At 30°C , the fifth C-V curves at 0.5 and 0.02 V/s appeared nearly equivalent. That is, the hole emission time at 30°C is much longer than the measurement time scales. In contrast, the significant change for sweep rates was observed at 200°C . By decreasing a sweep rate as well as increasing a measurement time, the negative shift in the fifth sweep was suppressed especially for the sweep rate of 0.02 V/s. This suggests enhanced hole emission at the higher temperature during repeated sweeps.

We introduced ΔV as a quantitative index. ΔV was defined as a difference between voltages at which a capacitance reached 80 nF/cm² in the first and fifth reverse sweeps. In this definition, ΔV indicates the negative voltage shift in the fifth reverse C-V curves relative to the first one; a larger ΔV corresponds to a larger negative shift. Figure 4(a) presents the reverse sweep segments extracted from repeated C-V measurements at 200°C . The different sweep rates of 0.5 , 0.05 , 0.02 , and 0.01 V/s correspond to a dual sweep period (i.e., sweep time from $+5$ to -15 to $+5$ V) of 80 , 844 , 2050 , and 4059 s, respectively.

Figure 4(b) plots ΔV against the sweep period. Here, we used the second sweep instead of the fifth to calculate ΔV for sweep times of 8078 and 12097 s to reduce the total measurement time because we found that at a sweep time of 4059 s, the second and fifth sweeps give almost identical results. In addition, for measurements at 225°C , the second sweep was also used to calculate ΔV for sweep times of 3054 and 4059 s. The obtained ΔV - t characteristics showed a clear trend: as the sweep period became longer, ΔV became smaller. At the same time, ΔV did not go to zero even at the longest sweep times we tested. Instead, it approached a certain value shown by a dotted line.

These results suggest that the traps near the valence band responsible for the observed Fermi-level pinning contained two components. One part was able to recover when the capacitor was allocated a longer period of time and high temperature, in which case

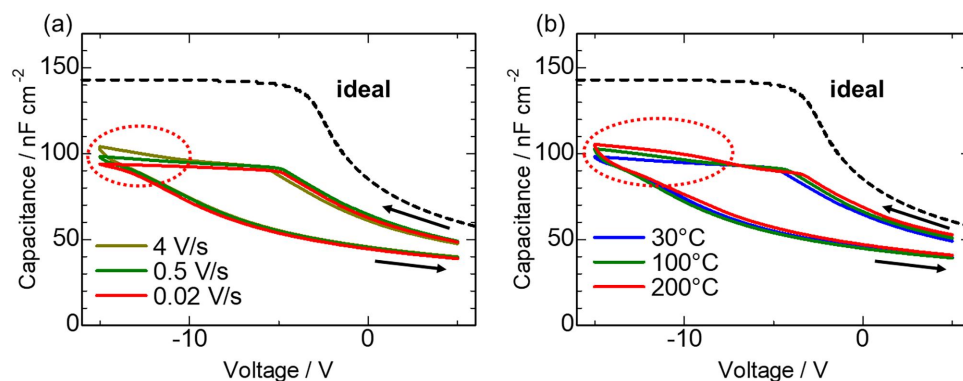


FIG. 2. (a) C-V characteristics measured at sweep rates of 4 , 0.5 , and 0.02 V/s at 30°C . (b) C-V characteristics with a sweep rate of 0.5 V/s measured at 30 , 100 , and 200°C . The accumulation capacitance was not reached due to strong Fermi-level pinning near the valence band edge.

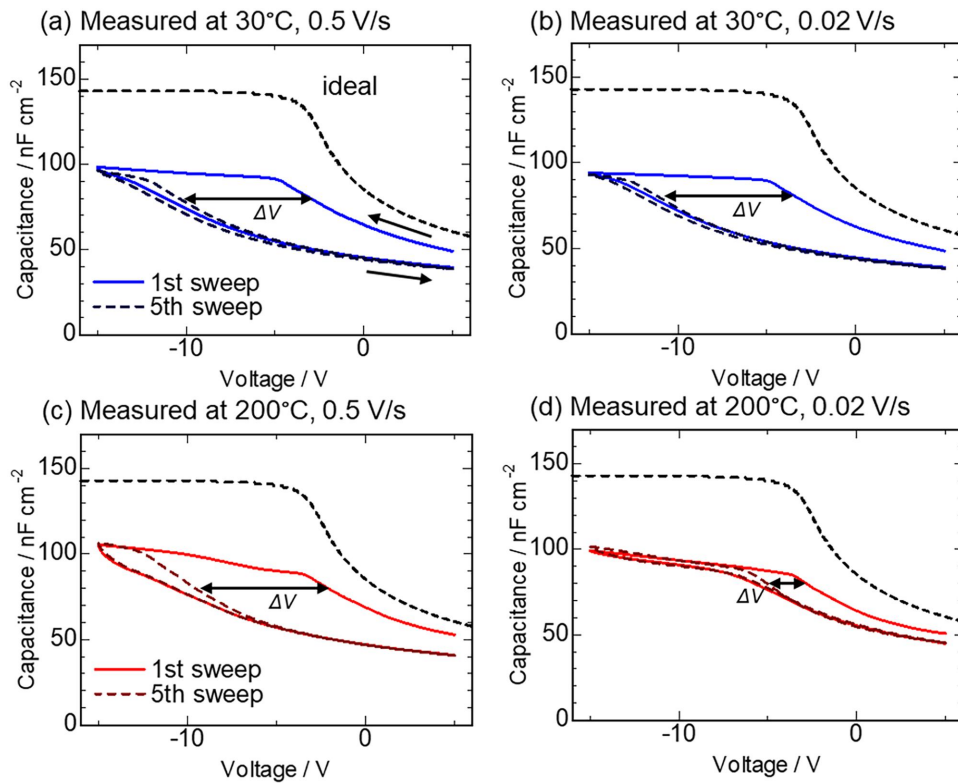


FIG. 3. First and fifth sweeps from a set of five consecutive C - V measurements. Measurement conditions were (a) 30 °C and 0.5 V/s, (b) 30 °C and 0.02 V/s, (c) 200 °C and 0.5 V/s, and (d) 200 °C and 0.02 V/s.

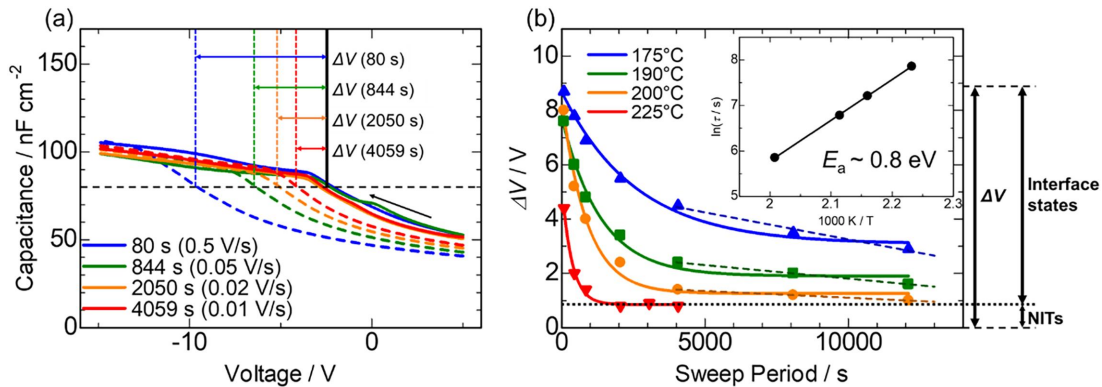


FIG. 4. (a) Reverse sweep segments extracted from repeated C - V measurements at 200 °C for the various sweep rates as well as the various measurement periods. The first sweep is shown as a solid line, and the fifth as a dashed line. (b) ΔV vs sweep period at 175, 190, 200, and 225 °C, with a single exponential fitting for all data points as shown by solid lines. Dashed lines are just guidelines in the sweep periods longer than 4059 s. Upper-right inset: Arrhenius plot of $\ln(\tau)$ vs $1000/T$ using τ from the ΔV - t analysis.

trapped holes were thermally emitted, and the negative shift decreased. The remaining parts did not recover under the same conditions. The presence of the traps caused ΔV in the ΔV - t characteristics to converge to a certain value.

We analyzed the ΔV - t data at each measurement temperature T by fitting it with a function,

$$\Delta V(t) = A \exp(-t/\tau(T)) + B, \quad (1)$$

where A is the initial shift amplitude, τ is the time constant as a function of T , and B is the residual shift at long times. Fitting curves almost agreed with the experimental data. In the sweep period longer than 4059 s at 175–200 °C, the change in ΔV looked like small. As indicated by the dashed guideline, we can confirm the slight decrease in ΔV even in long sweep periods. This trend suggests that the ΔV - t curves at 175–200 °C would also converge to the same value under extremely long measurement conditions. In contrast, ΔV values at

225 °C were rapidly decreased to the constant value even in the 2050 s. We extracted τ for each temperature from these fitting and then constructed an Arrhenius plot of $\ln(\tau)$ vs $1000\text{ K}/T$ as shown in the upper-right inset of Fig. 4(b). Assuming $\tau = \tau_0 \exp(E_a/k_B T)$, E_a was estimated to be approximately 0.8 eV.

This value of 0.8 eV exhibited reasonable agreement with the pinning level of approximately $E_V + 0.8\text{ eV}$ estimated from the surface potential at the capacitance plateau. This supports the interpretation that the slight decrease in the negative shift was caused by thermal emission from interface states with a deep energy level. On the other hand, we associated the certain value that the shift approached over long periods of time with NITs in the dielectric, which were rarely emitted thermally.

Figure 5 shows the defect model near the valence band maximum. The model consists of two components, including (i) interface states located at a depth of about 0.8 eV above the E_V of GaN and (ii) NITs in the dielectric. When the gate bias was swept from positive to negative, holes were captured in a high density of interface states, which caused Fermi-level pinning. A part of holes transferred to NITs from interface states. When the bias was swept back from negative to positive, trapped holes were not emitted at room temperature within measurement periods, resulting in a large negative shift. At high temperature and with sufficiently long sweep durations, holes captured in interface states were thermally emitted, which led to the decrease in the negative shift. In contrast, holes that transferred into NITs remained trapped even at high temperature, which was attributed to the remaining negative shift. The physical origin of the NITs has not yet been clarified and remains an important issue for future work. However, in MOS devices, border traps have often been discussed in general as being associated with oxygen vacancies.⁴⁰

In conclusion, our results demonstrate that the AlSiO/p-type GaN interface annealed at 400 °C PDA contained two types of traps. Interface states exhibited thermally activated hole emission at elevated temperatures, whereas NITs in the dielectric exhibited negligible thermal hole emission. Our analysis was based on temperature-dependent repeated C–V measurements with various sweep periods. By

analyzing the voltage shifts, we successfully separated trap components between interface states and NITs and estimated an energy level of interface states. These findings would provide direct insight into dominant factors of negative bias instability in GaN power MOSFETs. In addition, this approach should be useful not only for AlSiO/GaN systems but also for other wide-bandgap MOS interfaces in which slow carrier emission dominates bias instability.

See the [supplementary material](#) for C–V curves for five repeated dual sweeps at 30 and 200 °C for sweep rates of 0.5 and 0.02 V/s [Figs. S1(a)–S1(d)].

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

Author Contributions

Fumiyuki Sei: Conceptualization (lead); Data curation (lead); Formal analysis (lead); Investigation (equal); Methodology (equal); Validation (equal); Visualization (lead); Writing – original draft (lead); Writing – review & editing (lead). **Kenji Ito:** Conceptualization (equal); Formal analysis (equal); Investigation (equal); Resources (equal); Validation (equal); Writing – review & editing (supporting). **Tetsuo Narita:** Conceptualization (equal); Data curation (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Resources (equal); Validation (equal); Visualization (equal); Writing – review & editing (equal). **Masahiro Horita:** Conceptualization (equal); Investigation (equal); Project administration (equal); Resources (equal); Supervision (equal); Validation (equal); Writing – review & editing (equal). **Tetsu Kachi:** Conceptualization (equal); Data curation (equal); Formal analysis (equal); Investigation (equal); Methodology (equal); Project administration (equal); Resources (equal); Supervision (equal); Validation (equal); Visualization (equal); Writing – review & editing (equal). **Jun Suda:** Conceptualization (equal); Data curation (equal); Formal analysis (equal); Funding acquisition (lead); Investigation (equal); Methodology (equal); Project administration (lead); Resources (equal); Supervision (lead); Validation (equal); Visualization (equal); Writing – review & editing (equal).

DATA AVAILABILITY

The data that support the findings of this study are available from the corresponding author upon reasonable request.

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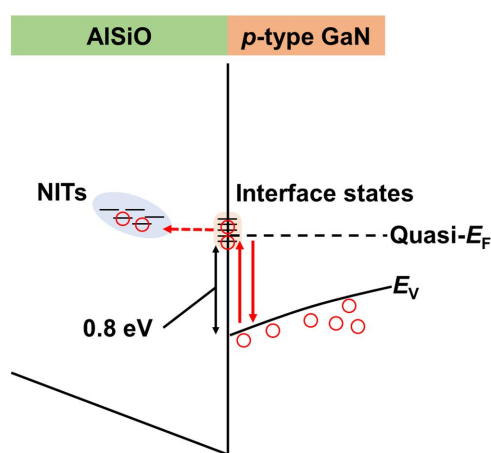


FIG. 5. Schematic band diagram with interface states and NITs near the valence band maximum at the AlSiO/p-type GaN interface. Fermi level (E_F) under negative bias was likely pinned at a high density of interface states.

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